

Section 3 Material properties

3.1 General

3.1.1 Strength and stiffness parameters

(1)P Strength and stiffness parameters shall be determined on the basis of tests for the types of action effects to which the material will be subjected in the structure, or on the basis of comparisons with similar timber species and grades or wood-based materials, or on well-established relations between the different properties.

3.1.2 Stress-strain relations

(1)P Since the characteristic values are determined on the assumption of a linear relation between stress and strain until failure, the strength verification of individual members shall also be based on such a linear relation.

(2) For members or parts of members subjected to compression, a non-linear relationship (elastic-plastic) may be used.

3.1.3 Strength modification factors for service classes and load-duration classes

(1) The values of the modification factor k_{mod} given in Table 3.1 should be used.

(2) If a load combination consists of actions belonging to different load-duration classes a value of k_{mod} should be chosen which corresponds to the action with the shortest duration, e.g. for a combination of dead load and a short-term load, a value of k_{mod} corresponding to the short-term load should be used.

3.1.4 Deformation modification factors for service classes

(1) The values of the deformation factors k_{def} given in Table 3.2 should be used.

3.2 Solid timber

(1)P Timber members shall comply with EN 14081-1. Timber members with round cross-section shall comply with EN 14544.

NOTE: Strength classes for timber are given in EN 338.

(2) The effect of member size on strength may be taken into account.

(3) For rectangular solid timber with a characteristic timber density $\rho_k \leq 700 \text{ kg/m}^3$, the reference depth in bending or width (maximum cross-sectional dimension) in tension is 150 mm. For depths in bending or widths in tension of solid timber less than 150 mm the characteristic values for $f_{\text{m},k}$ and $f_{\text{t},0,k}$ may be increased by the factor k_{h} , given by:

$$k_{\text{h}} = \min \left\{ \left(\frac{150}{h} \right)^{0,2}, 1,3 \right\} \quad (3.1)$$

where h is the depth for bending members or width for tension members, in mm.

Table 3.1 – Values of k_{mod}

Material	Standard	Service class	Load-duration class				
			Permanent action	Long term action	Medium term action	Short term action	Instantaneous action
Solid timber	EN 14081-1	1	0,60	0,70	0,80	0,90	1,10
		2	0,60	0,70	0,80	0,90	1,10
		3	0,50	0,55	0,65	0,70	0,90
Glued laminated timber	EN 14080	1	0,60	0,70	0,80	0,90	1,10
		2	0,60	0,70	0,80	0,90	1,10
		3	0,50	0,55	0,65	0,70	0,90
LVL	EN 14374, EN 14279	1	0,60	0,70	0,80	0,90	1,10
		2	0,60	0,70	0,80	0,90	1,10
		3	0,50	0,55	0,65	0,70	0,90
Plywood	EN 636 Part 1, Part 2, Part 3 Part 2, Part 3 Part 3	1	0,60	0,70	0,80	0,90	1,10
		2	0,60	0,70	0,80	0,90	1,10
		3	0,50	0,55	0,65	0,70	0,90
OSB	EN 300 OSB/2 OSB/3, OSB/4 OSB/3, OSB/4	1	0,30	0,45	0,65	0,85	1,10
		1	0,40	0,50	0,70	0,90	1,10
		2	0,30	0,40	0,55	0,70	0,90
Particle-board	EN 312 Part 4, Part 5 Part 5 Part 6, Part 7 Part 7	1	0,30	0,45	0,65	0,85	1,10
		2	0,20	0,30	0,45	0,60	0,80
		1	0,40	0,50	0,70	0,90	1,10
		2	0,30	0,40	0,55	0,70	0,90
Fibreboard, hard	EN 622-2 HB.LA, HB.HLA 1 or 2 HB.HLA1 or 2	1	0,30	0,45	0,65	0,85	1,10
		2	0,20	0,30	0,45	0,60	0,80
Fibreboard, medium	EN 622-3 MBH.LA1 or 2 MBH.HLS1 or 2 MBH.HLS1 or 2	1	0,20	0,40	0,60	0,80	1,10
		1	0,20	0,40	0,60	0,80	1,10
		2	–	–	–	0,45	0,80
Fibreboard, MDF	EN 622-5 MDF.LA, MDF.HLS MDF.HLS	1	0,20	0,40	0,60	0,80	1,10
		2	–	–	–	0,45	0,80

(4) For timber which is installed at or near its fibre saturation point, and which is likely to dry out under load, the values of k_{def} , given in Table 3.2, should be increased by 1,0.

(5)P Finger joints shall comply with EN 385.

3.3 Glued laminated timber

(1)P Glued laminated timber members shall comply with EN 14080.

NOTE: In EN 1194 values of strength and stiffness properties are given for glued laminated timber allocated to strength classes, see annex D (Informative).

(2) The effect of member size on strength may be taken into account.

(3) For rectangular glued laminated timber, the reference depth in bending or width in tension is 600 mm. For depths in bending or widths in tension of glued laminated timber less than 600 mm